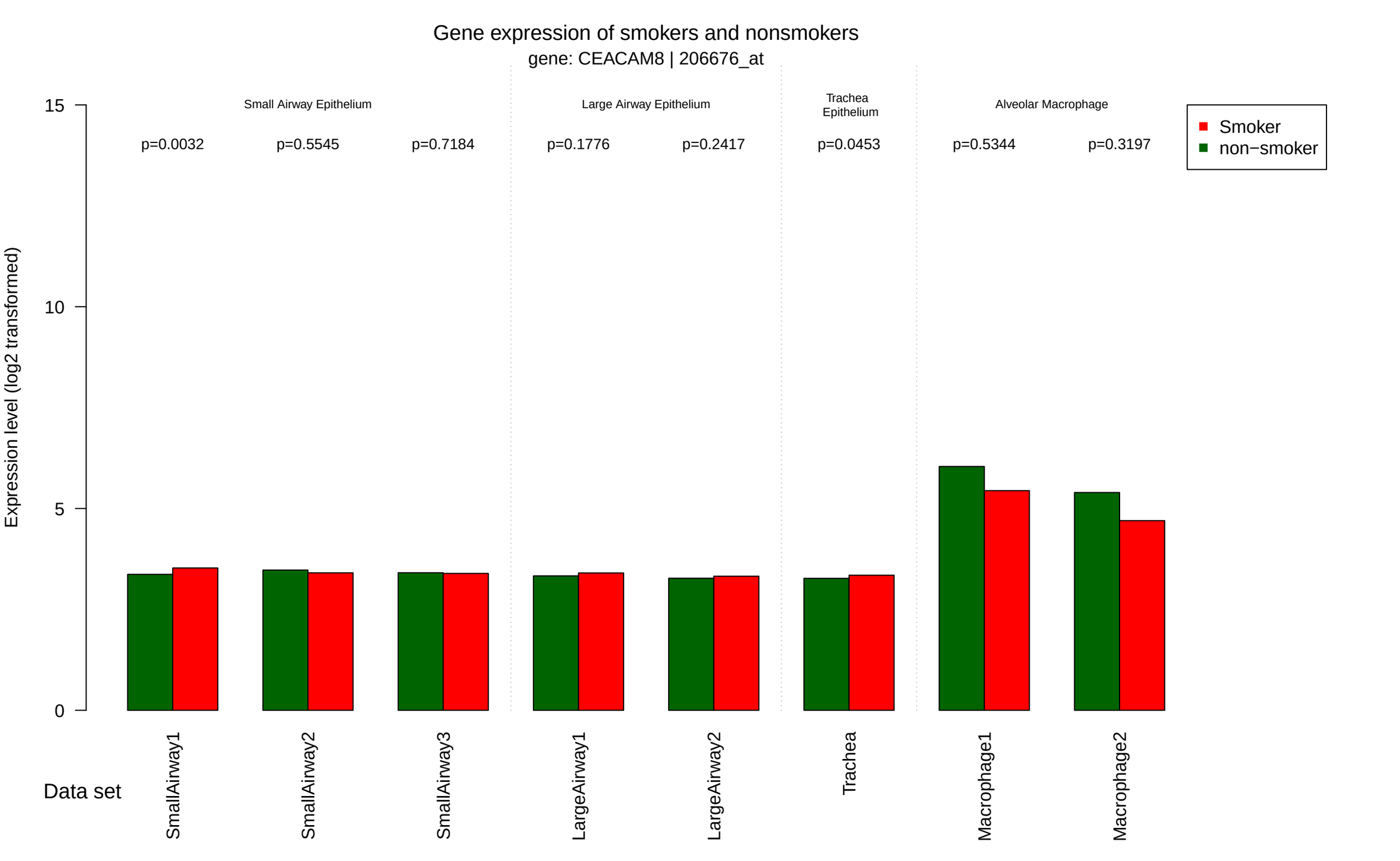
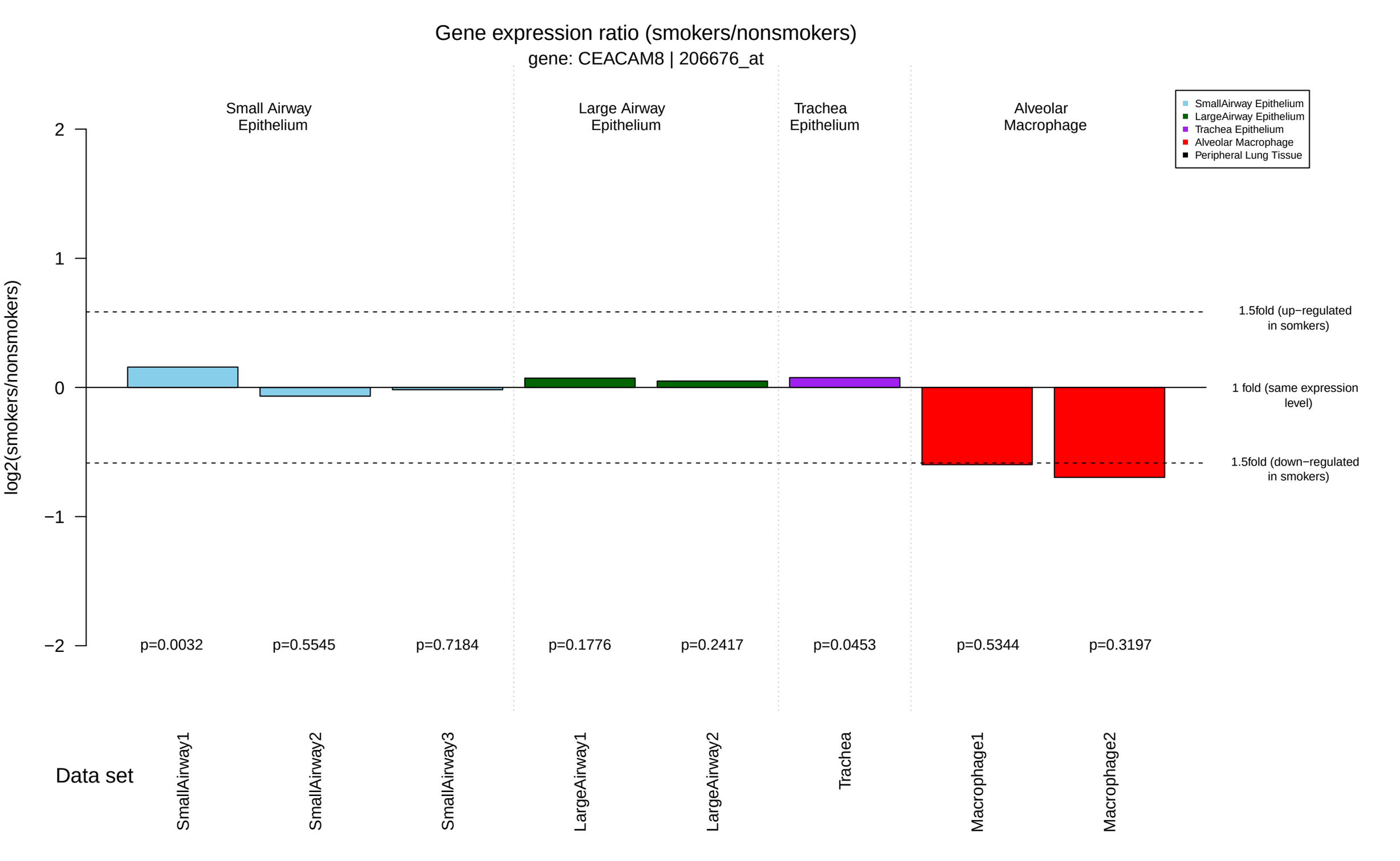
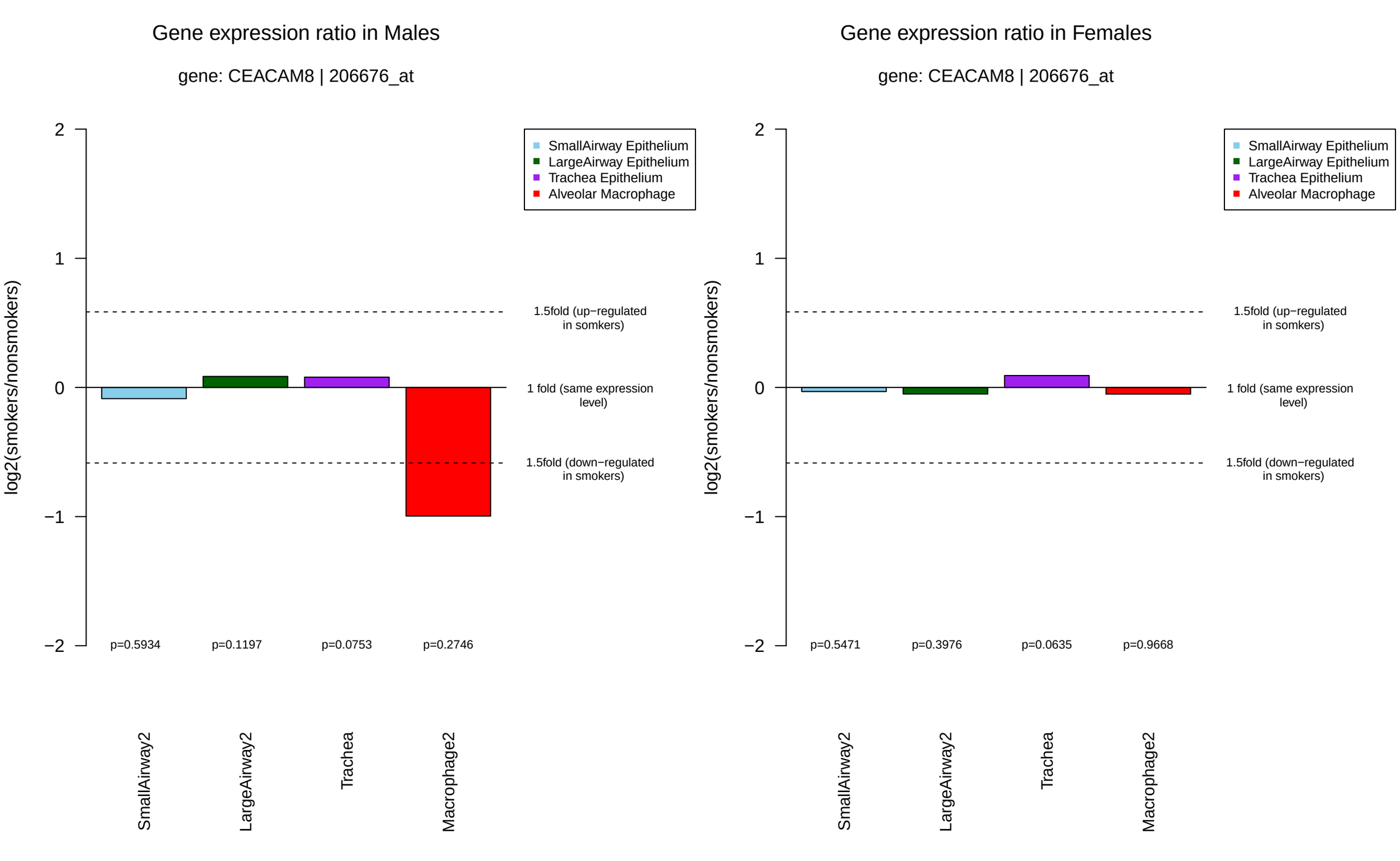
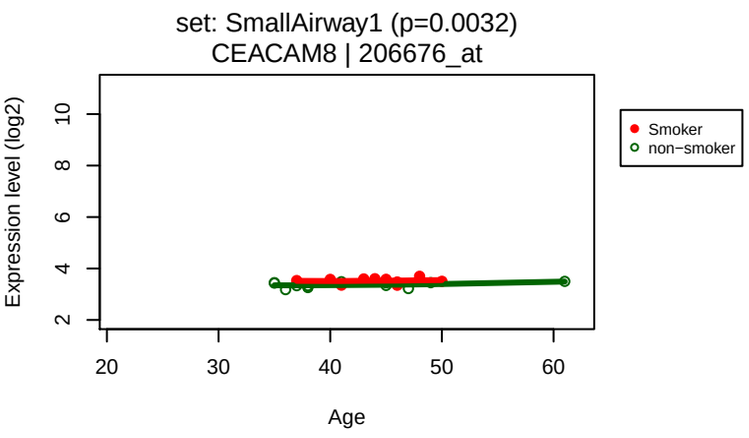
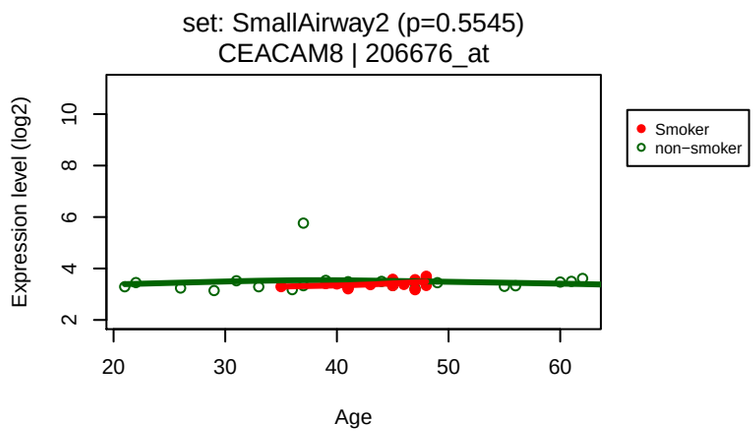
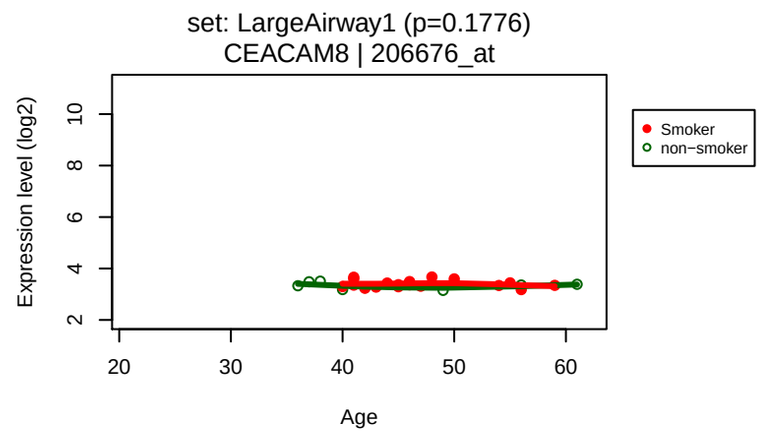

Gene expression of smokers and nonsmokers
gene: CEACAM8 | 206676_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0032
p=0.5545
p=0.7184
p=0.1776
p=0.2417
p=0.0453
p=0.5344
p=0.3197
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

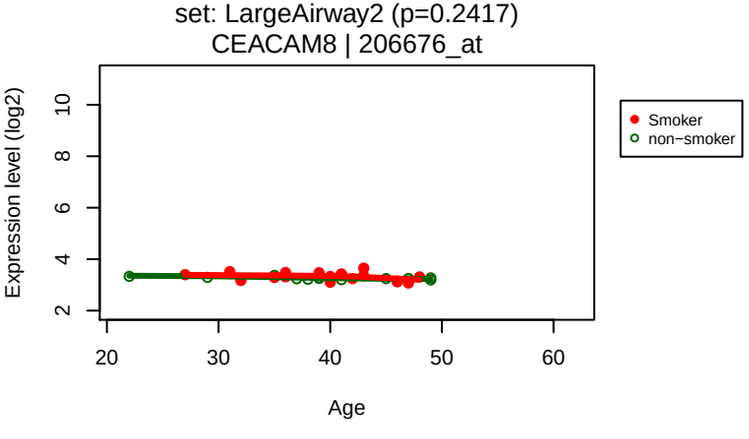

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM8 | 206676_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0032
p=0.5545
p=0.7184
p=0.1776
p=0.2417
p=0.0453
p=0.5344
p=0.3197
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

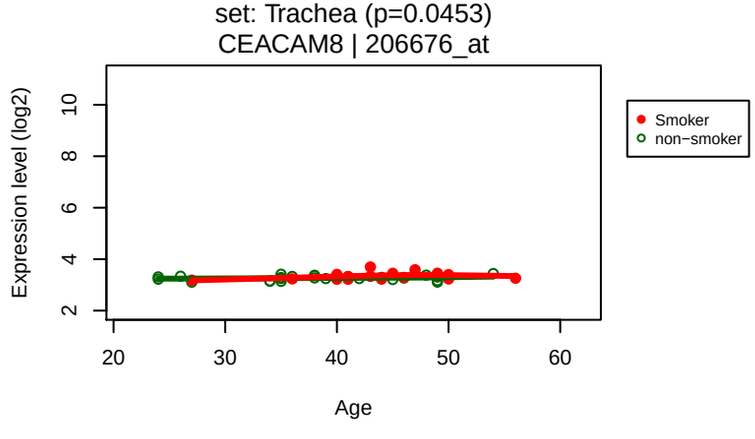

Gene expression ratio in Males
gene: CEACAM8 | 206676_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5934
p=0.1197
p=0.0753
p=0.2746
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM8 | 206676_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5471
p=0.3976
p=0.0635
p=0.9668
SmallAirway2
LargeAirway2
Trachea
Macrophage2

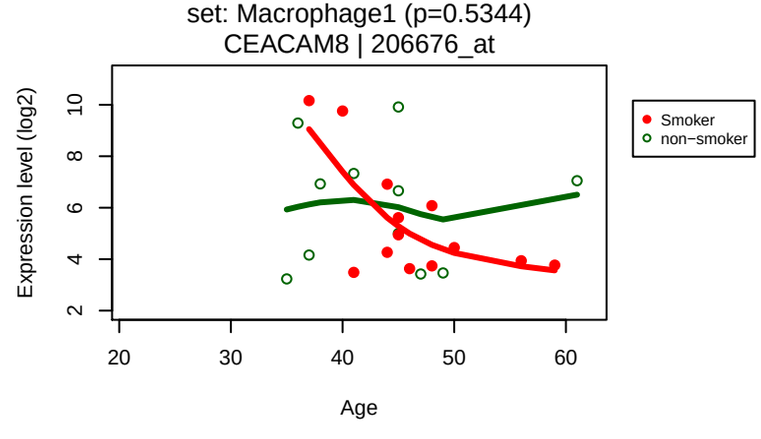

set: SmallAirway1 (p=0.0032)
CEACAM8 | 206676_at
Expression level (log2)
Age
Smoker
non−smoker

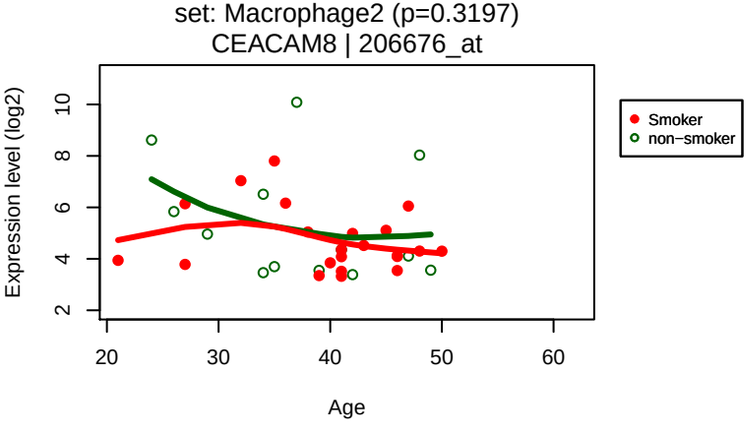

set: SmallAirway2 (p=0.5545)
CEACAM8 | 206676_at
Expression level (log2)
Age
Smoker
non−smoker

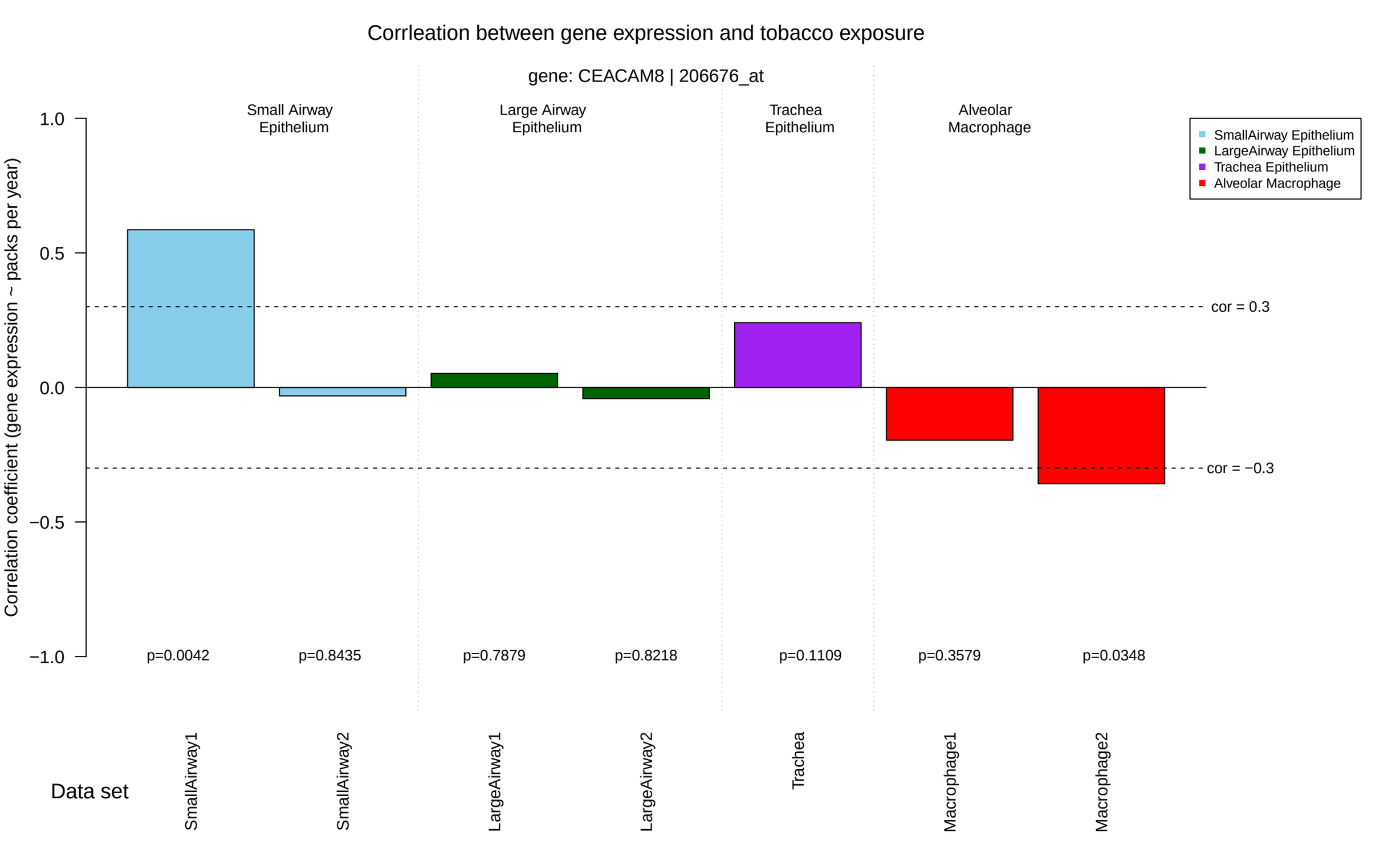

set: LargeAirway1 (p=0.1776)
CEACAM8 | 206676_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2417)
CEACAM8 | 206676_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0453)
CEACAM8 | 206676_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5344)
CEACAM8 | 206676_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3197)
CEACAM8 | 206676_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM8 | 206676_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0042
p=0.8435
p=0.7879
p=0.8218
p=0.1109
p=0.3579
p=0.0348
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2